Selected Undergraduate Competencies

1. **Critical Thinking and Problem Solving**
2. **Effective Oral and Written Communication**
3. **Collaboration Across Networks and Leading With Influence**

Competency 1: Critical Thinking and Problem Solving

- **Goal 1.1 (Research Focus):** By June 21, 2026, I will compile the evaluation metrics for the newly implemented K + d discrimination and triplet loss algorithms into a comprehensive results table. I will measure success by finalizing the data comparison across different SNRs and known/unknown splits, presenting it to my research supervisor to immediately slot into our paper draft.
- **Goal 1.2 (Workplace/Employer Goal):** Following a developmental conversation with my principal investigator for my Lassonde Undergraduate Research Award term, by July 15, 2026, I will optimize the anomaly detection threshold for the Unknown Signal Bank to decrease false positive rates by 10%. I will track this by analyzing the confusion matrices generated during the incremental training phase to ensure the purity of the candidate selection.

Why have you selected this Competency and its importance to your development?
Engineering is fundamentally about solving complex, high-stakes problems. My immediate tasks of finalizing data for an open-set modulation classification paper require rigorous analytical capabilities. This competency is vital to my development because it bridges theoretical design with practical execution. Refining my critical thinking allows me to build robust architectures, like integrating Extreme Value Theory for signal knownness, and optimize them for reliability in dynamic wireless environments. Mastering this ensures I can deliver tangible value before my research term concludes in August.

Competency 2: Effective Oral and Written Communication

- **Goal 2.1 (Academic Publishing):** By July 1, 2026, I will write the "Numerical Results and Discussions" section of our research paper, analyzing the stability of the Unknown Signal Bank and our dynamic adaptive threshold metrics. Progress will be measured by completing two iterative revision cycles with my principal investigator to ensure the manuscript is ready for submission before my term ends in August 2026.
- **Goal 2.2 (Mentorship Communication):** By November 15, 2026, I will design and deliver a midterm review presentation for the MATH 1028 students I mentor. I will evaluate my communication effectiveness by collecting anonymous surveys from the attendees, aiming for an 85% satisfaction rate regarding the clarity of my mathematical explanations.

Why have you selected this Competency and its importance to your development?
Whether drafting fast-tracked peer-reviewed publications or mentoring undergraduate students, clear communication is essential. Articulating complex research findings, such as how our hybrid DONet framework isolates unknown modulations from known distributions, is required to contribute meaningfully to the academic community. Furthermore, translating abstract mathematical concepts for MATH 1028 university students challenges me to tailor highly technical information to diverse audiences. Mastering this competency will allow me to successfully transition between the roles of a highly technical researcher and an accessible, supportive educator.

Competency 3: Collaboration Across Networks and Leading With Influence

- **Goal 3.1 (Mentorship Collaboration):** By October 30, 2026, upon returning to the university semester, I will partner with two other Peer Mentors to collaboratively develop and distribute a comprehensive mid-term study guide for MATH 1028. I will track resource utilization by counting the number of students who scan our shared QR codes or reply to our email distribution, ensuring at least 50 documented interactions.
- **Goal 3.2 (Research Teamwork):** By August 10, 2026, I will coordinate a final technical handover session with my research co-authors. I will measure this collaboration by successfully documenting the K + d clustering implementation and codebase architecture, ensuring the team can seamlessly continue modifying the DONet framework after my research term concludes.

Why have you selected this Competency and its importance to your development?
Innovation does not happen in a vacuum as it requires working efficiently within a team and taking proactive leadership. My upcoming transition out of my summer research role means I must effectively collaborate to document and hand off my contributions, ensuring my work on the metric-learning framework leaves a lasting impact. Furthermore, as I resume my Peer Mentor role in the fall, I want to lead by example, collaborating with peers to create scalable resources that support junior engineering students. Refining this competency ensures I can drive joint milestones and foster a supportive academic environment.